

D 131410

(Pages : 3)

Name.....

Reg. No.....

**THIRD SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2025**

Common Course

A11—BASIC NUMERICAL METHODS

(2020—2023 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A*All questions can be attended.**Each questions carries 2 marks.*

1. What do you mean by measures of central tendency ?
2. What is the degree of a quadratic equation ?
3. What are the different methods of solving a quadratic equation ?
4. What is square matrix ?
5. What is scalar matrix ?
6. What do you mean by geometric progression ?
7. Write the limitations of arithmetic progression.
8. What do you mean by skewness ?
9. Find compound interest on Rs. 20,000 for 2 years at 5 % p.a. compounded annually.
10. What do you mean by quartile deviation ?
11. Calculate range and its co-efficient from the following data :
12. 80, 85, 75, 74, 78, 95, 74
13. Calculate median : 52, 45, 65, 74, 78, 45, 78, 89
14. Calculate mode : 98, 74, 75, 72, 63, 62, 75, 99, 87, 72, 53, 72
15. Calculate mean : 12, 35, 65, 45, 14, 25, 35, 14, 25

(15 × 2 = 30 marks) Max. Ceiling : 25 marks

Turn over

Section B

*All questions can be attended.
Each questions carries 5 marks.*

16. From the following information find the value of standard deviation and its co-efficient :

Class	:	2	4	6	8	10	12
Frequency	:	12	11	12	13	14	12

17. What is present value of money ?

18. Find out harmonic mean.

Speed	:	50	100	125	150
Km. travelled	:	100	200	250	300

19. Distinguish between AP and GP.

20. Solve $5(x-1) + 2(x-2) + 15 = x - 7$.

21. Find out mode.

Class	:	2	4	8	12	10	20
Frequency	:	110	100	120	105	107	103

22. Find Q1 and Q2.

Class	:	5	10	15	18	12	13	15
Frequency	:	8	2	4	5	8	9	8

23. Find out median :

Class	:	10	20	30	40	50	60	70	80
Frequency	;	10	14	13	1	17	14	15	16

$(8 \times 5 = 40 \text{ marks})$ Max. Ceiling : 35 marks

Section C

*Answer any **two** questions.
Each questions carries 10 marks.*

24. Explain different measures of dispersion.
25. Solve the following by using Cramer's rule :

$$\begin{aligned}x + y - z &= 0 \\2x - y + z &= 0 \\3x + y + z &= 6.\end{aligned}$$

26. Find quartile deviation from the following data :

Classes	:	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Frequency	:	50	60	10	20	20	30	40

27. Compute mean, median and mode of the following data :—

Classes	:	20-25	25-30	30-35	35-40	45-50	50-55	55-60
Frequency	12 :	19	12	16	18	15	18	14

(2 × 10 = 20 marks)

D 131410–A**(Pages : 4)****Name.....****Reg. No.....****THIRD SEMESTER (CBCSS—U.G.) DEGREE EXAMINATION
NOVEMBER 2025****Common Course****A11—BASIC NUMERICAL METHODS****(2020–2023 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

A11—BASIC NUMERICAL METHODS

(Multiple Choice Questions for SDE Candidates)

1. One equation $ax^2 + 4 = 0$.
(A) Quadratic equation. (B) Cubic equation.
(C) Simple equation. (D) None of these.
2. One equation $4x^2 + 7 = 0$ is known as :
(A) Pure. (B) General.
(C) Simple. (D) None of these.
3. Find the common ratio of the following : 9, 6, 4
(A) 3. (B) 2.
(C) $\frac{2}{3}$. (D) None of these.
4. The value of determinant $\begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix}$ is :
(A) 0. (B) 6.
(C) - 6. (D) ± 6 .
5. A matrix in which every element is zero.
(A) Unit. (B) Diagonal.
(C) Scalar. (D) Null.
6. A series is obtained by adding a constant number to its preceding term is _____.
(A) G.P. (B) A.P.
(C) G.P. or A.P. (D) None.
7. Statistics does not study :
(A) Individuals. (B) Groups.
(C) Aggregates. (D) All of these.

8. The primary data are collected by
- (A) Interview. (B) Observation.
(C) Questionnaire. (D) All of these.
9. The classification 10 - 19, 20 - 29, 30 - 39 is the example of _____.
(A) Exclusive. (B) Inclusive.
(C) Both. (D) None.
10. The presence of extreme observation affects :
(A) A. M. (B) Median.
(C) Mode. (D) All.
11. _____ is called positional measure.
(A) Mean. (B) Media.
(C) Mode. (D) H.M.
12. The distribution for which C.V. is less, is _____ consistant
(A) More. (B) Less.
(C) Moderate. (D) None.
13. If mean is 100 and SD is 15 then c.v is _____.
(A) 66.67. (B) 100.
(C) 15. (D) None.
14. Mean of 3 items is 40. Two items are 30 and 50. What is the other ?
(A) 30. (B) 50.
(C) 40. (D) None.
15. In pie diagram, divisions are shown by means of _____.
(A) Circle. (B) Sector.
(C) Circle or Sector. (D) None.

Turn over

16. Frequencies of all the preceding classes are added to the frequency of a class, it is called———.
- (A) Frequency. (B) Class.
(C) More than cumulative. (D) Less than cumulative.
17. ——— is filled by the enumerator.
- (A) Questionnaire. (B) Schedule.
(C) Questionnaire or schedule. (D) All.
18. If the mid value of classes are 2.5, 7.5, 12.5, 17.5 the first class is :
- (A) 2.5 – 7.5 (B) 2.5 – 2.5.
(C) 5 – 10. (D) 0 – 5.
19. The process of arranging data in group according to similarities is called :
- (A) Tabulation. (B) Classification
(C) Tabulation or classification. (D) None.
20. Circle diagram is also called :
- (A) Pictogram. (B) Cartogram.
(C) Pie diagram. (D) None.